

CCST9012 Tutorial Week 3

Use of Armillary Sphere, Moon

In Lecture 4, we learned that the Armillary Sphere is a practical device that can be used to predict the motion of the sun any time of year. In Lecture 5 we learned about the relationship between the Moon's phases, two different lengths of a month. In this week's discussion, you will use the armillary sphere to track the sun in Hong Kong on your birthday and use an interactive animation to explore the Moon's motion in the ecliptic plane.

1 USE THE ARMILLARY SPHERE

We will use the Armillary Sphere app to track the motion of the Sun as it would be seen in Hong Kong on your birthday.

The virtual Armillary Sphere in the app is a deluxe model, in that the North Celestial Pole and Celestial equator can be oriented with respect to the horizon ring at different angles to accommodate predictions for observers at different latitudes.

Answers for this section will change depending on your birthday. For more insight into this section, check the slide, "How Good is the Armillary Sphere"

Select the Meridian Slider near the bottom of the screen and rotate the celestial sphere until the Latitude near the top of the screen reads 22.3 degrees.

To track the motion of the sun on the Armillary Sphere, the first thing we must do is place a mark on the ecliptic ring on your birthday.

Select the Ecliptic Slider and adjust it to your birthday. When you have done so, tap 'Mark'.

When you do this, the numbers representing the time of day, on the Celestial Equator ring will rotate such that 12 is intersecting the horizon. You have now located the position of the Sun at noon on your birthday. The altitude of the sun can be read at the top of the screen. On a real Armillary Sphere this can be done by noting the difference between the reading on the meridian ring where it intersect the horizon and where it intersects your mark on the ecliptic ring.

Note the noon altitude.

By selecting the equator slider, you can mimic the daily motion of the sun. As you do so, the time of day can be seen by which mark on the Celestial Equator ring intersects the Meridian. Rotate the celestial sphere until your mark representing the Sun is just below the horizon (Altitude = -0.8 degrees) in the East. You have now made a prediction of the time and location of sunrise on your birthday. On a real Armillary Sphere this can be done by reading the Azimuth off the horizon Ring. The app conveniently does this for you.

Note the sunrise time and Azimuth.

Repeat the procedure for Sunset.

Report your findings on the shared Google Spreadsheet.

2 THE MOON

Open the Interactive Moon applet.

This simulates the motion of the Sun and Moon in circular orbits around the Earth. The view is looking down from the North *Ecliptic* Pole and the plane of their orbits is the ecliptic plane. From the Earth, we see the Moon and Sun participate in the Daily (diurnal) motion of the celestial sphere, rising in the East and setting in the West. However, they

both have a second motion along the ecliptic from West to East. This simulation ignores the daily motion and shows only the second motion. The largest circle represents the ecliptic itself and the points 'Moon's Zodiac' and 'Sun's Zodiac' can be thought to represent which of the 12 zodiac constellations the Moon and Sun can be seen visiting.

2.1 MOON PHASES

In the lower left, the phase of the moon is shown along with a two-word description. It is either waxing, (getting larger) or waning (getting smaller). It is either a crescent, (less than $\frac{1}{2}$ illuminated), or a gibbous, (more than $\frac{1}{2}$ illuminated).

The angle between an object, the Earth and the Sun is called that object's elongation.

What is the elongation for a new moon?

The elongation of the new moon is zero degrees.

What is the elongation for a full moon?

The elongation of the full moon is 180 degrees.

In this simulation, the Moon's orbit has $\frac{1}{2}$ the radius as the sun's orbit. This is not realistic and creates some differences between the way this simulation illustrates the phases of the moon and Figure 8.3 in our textbook, which assumes that the Sun is infinitely far away from the Earth. In Chapter 12, we will learn that an attempt to measure the elongation of the quarter moon is the first recorded attempt at estimating the distance between the Earth and the Sun.

In reality, the elongation of a quarter moon is just barely less than 90 degrees. In this simulation, what is the elongation for a quarter moon?

In this simulation, the elongation of the quarter moon is only 60degrees. This angle depends on the relative size of the circle of Moon's orbit and the circle of Sun's Orbit. It is not possible to clearly show these to scale in a small computer screen.

Elongation also tells us the difference between the time of sunrise and moonrise.

Consider the daily motion of the celestial sphere. If the elongation of the moon is 90 degrees, how much of a day will separate sunrise and moonrise?

We follow the usual practice of freezing the annual motion of the sun and second motion of the moon when we consider the daily motion of the celestial sphere. Their locations on the celestial sphere will be 90 degrees from each other and thus $\frac{1}{4}$ of a rotation of the celestial sphere will separate sunrise and moonrise. $\frac{1}{4}$ of a day separates them.

2.2 TWO LENGTHS FOR A MONTH

In Lecture 5, we learned that there are two definitions for a Month. A *synodic month* refers to the time from one full moon to the next. A *sidereal month* refers to the time it takes the moon to visit all 12 zodiac constellations.

Move the Moon Position and Sun Position sliders to zero degrees. Start the animation box. Stop it when the moon is new again. Has the moon completed 1 circle, less than 1 circle or more than 1 circle? Which is longer, the sidereal month or the sidereal month?

The moon completes more than one circle from one new moon to the next new moon. This means the sidereal month, (the time it takes for the moon to complete one circle) is shorter than the synodic month.

The speed of the Sun's annual motion can be changed by the year length slider. This is measured in terms of months. Which definition of month?

To answer this question, one must check if the time it takes to complete one circle is equal to the number of new moons it makes in that time, or equal to the number of circles the moon makes. With the default setting of 13.4, it would take a long time to check. If you set the year length to 2, you can see that the sun makes one circle in the same time it takes the sun to make 2 circles. It is measured in Sidereal months.

Can you find a value of Year Length such that the moon does not change its phase?

Yes. If the value of Year Length is 1, then the Sun and Moon pass through the zodiac constellations at the same speed and their phase remains constant.